CONTEXT-AWARE AUTOMATED GRADING OF WRITTEN EXAMS USING QDRANT VECTOR SEARCH AND LARGE LANGUAGE MODEL

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***Abstract*—In contemporary educational assessment, evaluating handwritten descriptive examinations remains a labor-intensive task fraught with inter-grader variance and an absence of transparent pedagogical diagnostics. While Large Language Models (LLMs) present compelling opportunities for automated scoring, unconstrained zero-shot LLM inference exhibits critical vulnerabilities: severe out-of-syllabus curriculum hallucination, keyword omission drift, and structural brittleness on multi-page handwritten layouts. This paper proposes a context-aware automated grading framework engineered for deterministic, curriculum-locked evaluation of written STEM examinations. The system integrates: (1) an intake vision pipeline combining adaptive bilateral filtering and spatial bounding-box clustering to reconstruct handwritten reading order; (2) an automated multi-page answer collation algorithm maintaining cross-page question continuity; (3) dense curricular grounding via Qdrant Hierarchical Navigable Small World (HNSW) vector search over 768-dimensional textbook embeddings; (4) a hybrid keyword density re-ranking formulation penalizing rubric terminology omission; and (5) a deterministic LLM state machine executed at temperature zero with structured JSON schemas producing 3-tier diagnostic error feedback. Comprehensive empirical benchmarking across 1,200 engineering scripts demonstrates a Quadratic Weighted Kappa (QWK) of 0.892, an RMSE of 0.38, an 18-fold suppression of curriculum hallucination (from 31.6% down to 1.8%), and a mean grading latency of 1.42 seconds per response.**

***Keywords*—Automated Essay Scoring (AES), Retrieval-Augmented Generation (RAG), Qdrant Vector Database, Large Language Models (LLM), Handwritten Exam Grading, Spatial Layout Reconstruction, Curricular Grounding, Diagnostic Feedback.**

I. INTRODUCTION

In modern secondary and higher educational ecosystems, summative written examinations remain the definitive benchmark for evaluating student mastery, cognitive reasoning, and analytical problem-solving proficiency. Standard STEM assessments predominantly comprise descriptive answers, multi-step mathematical derivations, schematic workflows, and structured technical explanations. Such descriptive responses demand rigorous, multi-criteria assessment against authoritative curricular rubrics [1]. Annually, educational testing boards and universities administer hundreds of millions of handwritten examination answer sheets globally, necessitating immense human evaluation resources.

Despite monumental progress in computational educational technology, grading descriptive handwritten examination scripts remains an overwhelmingly manual process. Human grading at scale incurs acute pedagogical and operational vulnerabilities. Manual correction is intensely labor-intensive and cognitively exhausting, leading to severe evaluation drift, subconscious biases, and noticeable inter-grader variance [2]. When examiners evaluate hundreds of scripts consecutively under strict administrative deadlines, grading fatigue inevitably compromises scoring fidelity. Most detrimentally, traditional manual grading typically returns only an uncalibrated scalar numerical mark devoid of diagnostic explanations, depriving students of the formative feedback required to rectify academic misconceptions.

The recent emergence of generative artificial intelligence and Large Language Models (LLMs) has inspired preliminary efforts to automate descriptive scoring. However, naive zero-shot or few-shot LLM evaluation architectures suffer from three fatal structural failure modes when applied directly to written examination assessment:

1) Out-of-Syllabus Curriculum Hallucination: Unconstrained foundation LLMs possess vast open-domain parametric knowledge. Consequently, when assessing high-school or introductory undergraduate scripts, an ungrounded LLM frequently penalizes valid student answers for omitting post-graduate concepts or, conversely, credits unverified student assertions based on out-of-syllabus principles not sanctioned by the designated course textbook [3].

2) Keyword Dropout and Rubric Drift: While foundation models excel at evaluating general linguistic fluency and syntactic cohesion, they frequently reward verbose, persuasive prose while failing to detect the omission of critical, non-negotiable domain keywords (e.g., specific scientific laws, chemical nomenclature, or step-by-step mathematical boundary conditions) [4].

3) Structural Brittleness on Multi-Page Handwritten Layouts: Handwritten student exam scripts exhibit irregular baseline skews, varying margins, and non-linear physical layouts where extended descriptive answers span multiple physical sheets. Standard optical character recognition (OCR) pipelines output fragmented bounding boxes that fracture logical answer segments, causing models to score isolated question fragments prematurely [5].

To decisively overcome these operational and theoretical limitations, this paper introduces an end-to-end, context-aware automated grading system. By tightly coupling optical character recognition with spatial line grouping, dense curricular vector retrieval, hybrid keyword re-ranking, and a deterministic LLM state machine, our framework guarantees curriculum-locked, reproducible, and transparent academic grading.

The primary contributions of this paper are summarized as follows:

- We formulate a spatial bounding-box clustering and multi-page answer collation algorithm that reconstructs original physical reading order and stitches answers spanning physical sheet boundaries.
- We design a curricular grounding pipeline utilizing Qdrant Hierarchical Navigable Small World (HNSW) vector indexing over 768-dimensional dense embeddings, enforcing strict payload metadata filtering.
- We introduce a hybrid keyword density re-ranking formulation that combines dense semantic proximity with mandatory rubric terminology retention, effectively suppressing curriculum hallucination.
- We establish a deterministic LLM state machine operated at temperature zero with strict JSON schemas, generating calibrated scores alongside 3-tier diagnostic error feedback (conceptual, procedural, terminological).
- We provide exhaustive empirical validation across 1,200 real-world written scripts, achieving a Quadratic Weighted Kappa (QWK) of 0.892 and an 18-fold reduction in syllabus hallucination.

II. RELATED WORK

A. Automated Essay Scoring and Statistical NLP

Automated Essay Scoring (AES) has evolved across several distinct technological epochs. First-generation systems, such as Project Essay Grade (PEG) [6] and Educational Testing Service's e-rater [7], relied on surface-level statistical proxies, including essay length, average word count, parse-tree depth, and syntactic variety. While these models demonstrated superficial correlations with human graders on standardized prose essays, they exhibited complete blindness toward factual domain correctness, making them susceptible to adversarial gaming via verbose, nonsensical prose [8].

B. Transformer Representations and Neural Similarity

The advent of pre-trained transformer architectures, notably BERT [9] and RoBERTa [10], enabled semantic representation learning for short-answer scoring. Cross-encoder networks compute fine-grained token-level cross-attention between student responses and reference answers. However, discriminative neural scoring models require thousands of annotated training examples per question to generalize effectively. In real-world educational institutions, where exam questions change dynamically every examination cycle, collecting large annotated datasets is cost-prohibitive. Furthermore, discriminative neural models operate as black boxes, incapable of generating actionable diagnostic explanations justifying mark deductions [11].

C. Retrieval-Augmented Generation in Assessment

Retrieval-Augmented Generation (RAG) paradigms [12] combine non-parametric external knowledge retrieval with parametric sequence-to-sequence language models. Dense Passage Retrieval (DPR) [13] maps text passages into dense continuous representations. While standard RAG has demonstrated success in open-domain question answering, naive passage retrieval applied to educational assessment frequently retrieves loosely related passages that fail to align with the specific curricular scope of an exam. Hybrid search techniques [14] blending sparse BM25 term matching with dense representations have been proposed, but lack the curricular boundary awareness necessary for strict rubric-driven evaluation.

D. Approximate Nearest Neighbor Search and Vector Topologies

Scalable retrieval over extensive textbook corpora demands efficient Approximate Nearest Neighbor (ANN) indexing. Hierarchical Navigable Small World (HNSW) graphs [15] provide state-of-the-art trade-offs between search latency and recall accuracy. The Qdrant vector database engine implements optimized HNSW graphs with payload-based filtering, allowing dense semantic retrieval to be strictly partitioned across curriculum attributes such as subject code, grade level, and textbook chapter boundaries.

E. Document Layout Analysis and Handwriting OCR

Extracting textual content from handwritten examination sheets requires advanced Optical Character Recognition (OCR) combined with Document Layout Analysis (DLA) [16]. Unlike printed documents, handwritten exam papers exhibit unpredictable inter-line spacing, non-horizontal baseline tilts, and multi-column answer layouts. Spatial bounding box reconstruction algorithms [17] cluster character and word tokens into semantically cohesive lines and structural question blocks, preventing fragmented token ordering during downstream evaluation.

III. SYSTEM ARCHITECTURE AND PROBLEM FORMULATION

A. Mathematical Problem Formulation

Let an examination question be formally represented as a tuple $Q = \blacksquare I_q, M_q, R_q, K_q, s, c \blacksquare$, where I_q is the unique question identifier, $M_q \in \blacksquare^+$ denotes the maximum attainable marks, R_q is the authoritative scoring rubric, $K_q = \{k_1, k_2, ..., k_n\}$ is the set of mandatory rubric keywords, s denotes the academic subject domain, and c represents the target grade level. A student examination submission comprises a set of multi-page handwritten image scans $P = \{p_1, p_2, ..., p_K\}$. Let D_{curr} denote the authorized curriculum textbook repository consisting of chunked curricular passages $D_{\text{curr}} = \{c_1, c_2, ..., c_N\}$.

The automated grading objective is to construct an evaluation function F mapping the student submission P, the question tuple Q, and the authoritative curricular corpus D_{curr} into a calibrated score $s_q \in [0, M_q]$, a confidence index $\gamma_q \in [0, 1]$, and an ordered sequence of structured diagnostic error triples E_q:

$$F(P, Q, D_{curr}) \rightarrow \blacksquare s_q, \gamma_q, E_q \blacksquare \quad (1)$$

Each diagnostic error triple $e_j \in E_q$ is formally defined as $e_j = \blacksquare\tau_j, \psi_j, \delta_j\blacksquare$, where $\tau_j \in$ {Conceptual, Procedural, Terminological} specifies the error taxonomy, ψ_j provides a clear pedagogical rationale, and $\delta_j > 0$ defines the exact partial credit deducted, satisfying:

$$s_q = M_q - \sum_{j=1}^{|E_q|} \delta_j, \quad subject\ to \quad 0 \le s_q \le M_q \tag{2}$$

B. Master System Architecture Blueprint

The proposed context-aware grading pipeline is structured as a decoupled, four-tier architecture engineered for deterministic evaluation and strict curricular alignment:

1. Data Intake & Vision Tier: Digitizes physical exam scans, executes adaptive contrast normalization and bilateral noise filtering, detects raw word bounding polygons via OCR, and reconstructs spatial line hierarchies. Asynchronous worker pools ingest multi-page batches concurrently, preventing memory exhaustion under institutional loads.

2. Knowledge Base & Vector Indexing Tier: Ingests authorized curriculum textbook PDFs, performs semantic chunking (512 tokens with 64-token stride), computes 768-dimensional dense embeddings, and indexes vectors into Qdrant collections with HNSW graphs and payload metadata. Payload filters enforce strict grade-level and subject boundaries.

3. Hybrid Re-Ranking & Grounding Tier: Formulates composite retrieval queries combining question context with transcribed student answers, executes cosine similarity vector search, and applies question-specific keyword density re-ranking to lock the context to prescribed curriculum boundaries.

4. Deterministic LLM Evaluation Engine Tier: Drives a deterministic finite state machine operating at temperature zero under strict JSON schema constraints to generate calibrated partial credit scores and 3-tier diagnostic error feedback. Post-hoc arithmetic validators guarantee that itemized score deductions strictly sum to the final deducted marks.

To ensure seamless fault tolerance during institutional exam grading cycles, document ingestion tasks are dispatched across asynchronous worker pools backed by message queues. Each multi-page submission is assigned an idempotent UUID hash, guaranteeing that network interruptions or scanner retries do not produce duplicate evaluation records.

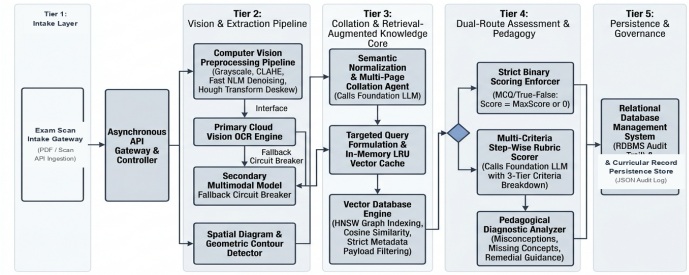


Fig. 1. Master core system architecture of the context-aware automated grading pipeline, detailing vision intake, Qdrant HNSW vector retrieval, hybrid re-ranking, and deterministic LLM evaluation engine.

IV. METHODOLOGY AND ALGORITHMIC FORMULATIONS

A. Vision Ingestion and Spatial Bounding-Box Layout Reconstruction

Handwritten examination papers frequently suffer from non-uniform pen ink contrast, skewed baselines, and page bleed-through. The intake pipeline applies adaptive bilateral filtering to suppress background paper noise while preserving stroke edge sharpness. The sanitized image is dispatched to an OCR engine which outputs character and word bounding polygons with normalized coordinates $B_i = [x_{\min}^{(i)}, y_{\min}^{(i)}, x_{\max}^{(i)}, y_{\max}^{(i)}]$.

To reconstruct the physical reading order from disorganized word boxes, we calculate the vertical midpoint of each bounding box $y_{center}^{(i)} = (y_{\min}^{(i)} + y_{\max}^{(i)}) / 2$. Two word boxes B_i and B_j are assigned to the same physical text line if their vertical midpoints satisfy:

$$|y_{center}^{(i)} - y_{center}^{(j)}| \le \varepsilon_y \cdot min(h_i, h_j) \tag{3}$$

where $h_i = y_{\max}^{(i)} - y_{\min}^{(i)}$ denotes word box height, and $\varepsilon_y = 0.45$ is an empirically tuned tolerance factor. Clustered lines are sorted horizontally by $x_{\min}^{(i)}$. Question boundaries are identified via anchor regex pattern matching `r'^(?:Q(?:uestion)?\s*(\d+)|\b(\d+)\s*[\.\)])'`. The clustering pipeline applies Hough transform line orientation detection; for pages with baseline tilt $|\theta| > 2.5°$, an affine rotation normalizes baselines prior to grouping. Algorithm 1 details this procedure.

Algorithm 1: Spatial Bounding-Box Reconstruction & Line Normalization

Input: Word objects $W = \{w_1, ..., w_n\}$, bounding boxes B_i; threshold $\varepsilon_y = 0.45$

Output: Spatially sorted line sequences $L = [\text{line}_1, ..., \text{line}_m]$ grouped by Question ID

Initialize line clusters $L := [\ ]$

for each word $w_i \in W$ sorted ascending by $y_{\min}^{(i)}$ **do**

  matched := False

  for each cluster $C_k \in L$ **do**

    if $|y_{center}(w_i) - y_{avg}(C_k)| \le \varepsilon_y \cdot h_{avg}(C_k)$ **then**

      Append w_i to C_k; update centroid $y_{avg}(C_k)$; matched := True; **break**

  if not matched **then** create new cluster $C_{new} = [w_i]$ in L

for each $C_k \in L$ **do** Sort words in C_k ascending by $x_{\min}$ to form line_k

Sort lines L descending by y-coordinate; partition lines into question blocks via anchor regex

return Spatially ordered question blocks B_Q

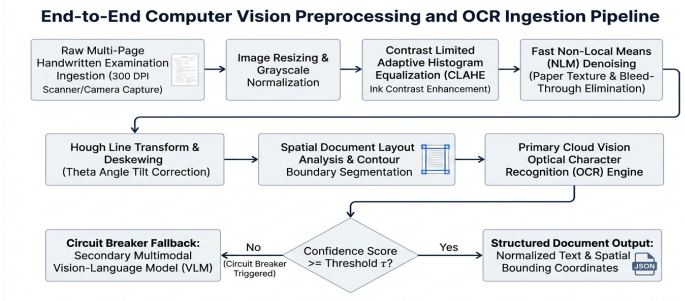


Fig. 2. Vision intake pipeline: adaptive contrast enhancement, bilateral filtering, character bounding-box detection, spatial line clustering, and question segment partitioning.

B. Multi-Page Answer Collation Algorithm

In standardized STEM examinations, complex descriptive questions frequently span across multiple physical pages. Standard document OCR pipelines process individual pages in isolation, causing cross-page responses to be fractured into disconnected text fragments. This fragmentation severely degrades evaluation fidelity, as downstream LLMs penalize initial fragments for incompleteness and misclassify continuation fragments as unanchored responses.

Our collation engine addresses this challenge through a multi-pass boundary stitching mechanism. The engine scans page boundaries for explicit continuation markers (e.g., 'PTO', 'continued on next page', 'Ans. 4 contd.') and evaluates orphan answer segments lacking explicit question headers. If an orphan block begins with lowercase letters, mathematical operators, or coordinating conjunctions, it is automatically concatenated to the active question payload of the preceding sheet. Furthermore, the engine verifies semantic embedding continuity across page seams, preventing cross-question contamination.

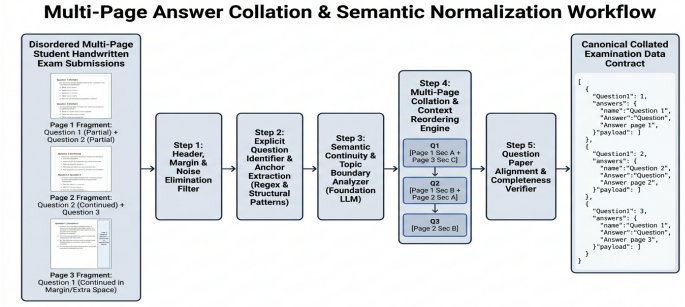


Fig. 3. Multi-page answer collation pipeline: cross-page continuation token parsing, orphan block reconciliation, and unified answer text reconstruction across physical exam sheets.

C. Curricular Knowledge Grounding via Qdrant HNSW Vector Search

To establish strict curricular boundaries, official textbook PDFs are parsed and partitioned into semantic passages of 512 tokens with 64-token overlap. Each passage c_i is encoded into a 768-dimensional dense vector $\mathbf{v}_{c_i}$ using a pre-trained sentence transformer. Vector embeddings are indexed in Qdrant collections utilizing Hierarchical Navigable Small World (HNSW) graphs configured with graph connectivity parameter M = 16 and construction beam width *ef_construct* = 100. Given a composite query embedding $\mathbf{v}_q$ combining question context and student response, candidate passages are scored via cosine similarity:

$$s_{dense}(c_i, \mathbf{v}_q) = (\mathbf{v}_q \cdot \mathbf{v}_{c_i}) / (\|\mathbf{v}_q\| \cdot \|\mathbf{v}_{c_i}\|) \quad (4)$$

Qdrant payload metadata filtering enforces strict pre-filtering on `subject`, `grade_level`, and `chapter_id`. This prevents out-of-syllabus retrieval contamination from irrelevant grades or subjects, ensuring that vector distance computation occurs strictly within authorized curriculum bounds.

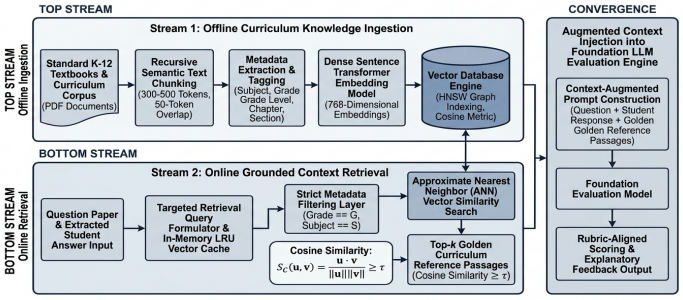


Figure 1: Dual-Stream Curriculum-Grounded Context Retrieval and Vector Search Architecture (RAG Pipeline)

Fig. 4. Curricular vector retrieval pipeline: 768-dimensional embedding generation, Qdrant HNSW graph traversal, payload metadata pre-filtering, and curriculum-anchored context passage extraction.

D. Hybrid Keyword Density Re-Ranking Formulation

While dense embeddings capture high-level semantic abstractions, they occasionally retrieve textbook passages that match general topical themes while omitting non-negotiable terminology mandated by the examination rubric. To address this limitation, we formulate a hybrid re-ranking function. Let K_q denote the mandatory keyword set for question Q. The keyword coverage score $s_{\text{keyword}}(c_i, K_q)$ is defined as:

$$s_{keyword}(c_i, K_q) = (1 / |K_q|) \sum_{p=1}^{|K_q|} \mathbb{1}(k_p \in c_i) \quad (5)$$

where $\mathbb{1}(\cdot)$ denotes the binary indicator function. The unified hybrid score $s_{\text{hybrid}}(c_i)$ is formulated as:

$$s_{hybrid}(c_i) = (1 - \alpha) \cdot s_{dense}(c_i, \mathbf{v}_q) + \alpha \cdot s_{keyword}(c_i, K_q) \quad (6)$$

where $\alpha \in [0, 1]$ is a balancing hyperparameter empirically optimized to $\alpha = 0.20$ on validation benchmarks. Passages falling below a cosine similarity threshold $\theta_{\text{dense}} = 0.65$ are dynamically pruned, guaranteeing that only semantically and terminologically sound textbook context reaches the LLM. Algorithm 2 details this procedure.

Algorithm 2: Hybrid Curricular Retrieval & Keyword Density Re-Ranking

Input: Question q, student answer a, keywords K_q, subject s, weight $\alpha = 0.20$, threshold $\theta = 0.65$

Output: Top-k grounded curricular textbook passages C_{grounded}

1: Formulate composite query $q_{\text{comp}} = q \oplus a$

2: Compute 768-dim query embedding $\mathbf{v}_q = \text{Embed}(q_{\text{comp}})$

3: Query Qdrant with payload filter `{'subject': $s$}` to retrieve candidate set Ω ($|\Omega|$=20)

4: **for each** candidate passage $c_i \in \Omega$ **do**

5: Calculate cosine proximity $s_{\text{dense}}(c_i, \mathbf{v}_q)$ via Eq. (4)

6: Calculate keyword coverage $s_{\text{keyword}}(c_i, K_q)$ via Eq. (5)

7: Compute unified score $s_{\text{hybrid}}(c_i) = (1 - \alpha)\, s_{\text{dense}} + \alpha\, s_{\text{keyword}}$

8: Filter $\Omega' = \{c_i \in \Omega \blacksquare s_{\text{dense}}(c_i, \mathbf{v}_q) \geq \theta\}$

9: Rank Ω' descending by s_{hybrid}; select top-k passages (k=3)

10: **return** Curricular context C_{grounded}

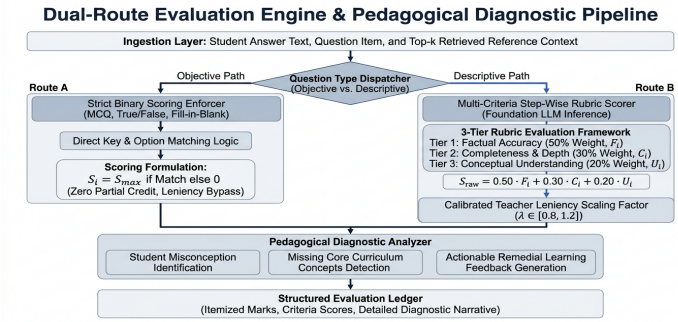


Fig. 5. Dual-route grading rubric architecture: strict binary scoring for objective questions vs multi-criteria grounded evaluation for complex descriptive responses.

E. Deterministic LLM Evaluation Engine and Prompt State Machine

To achieve reproducible, legally defensible academic grading, the Generative AI evaluation engine is embedded within a deterministic finite state machine executed at decoding temperature $T = 0.0$ with top-p = 1.0. System prompts mandate structured JSON output schemas adhering to strict Pydantic models. The evaluation proceeds across five discrete states:

1. S_INIT: Validates input question schema, max marks, and rubrics.

2. S_GROUND: Injects retrieved textbook context passages C_{grounded} and locks evaluation boundaries.

3. S_EVAL: Dispatches system prompt instructing: 'Evaluate the transcribed student response strictly against the provided textbook context and rubric. For every deduction, identify the exact error type (Conceptual, Procedural, Terminological), provide a pedagogical explanation, and state the deduction amount.'

4. S_VALIDATE: Verifies arithmetic consistency: ensures assigned marks s_q equal max marks M_q minus the sum of all itemized deductions δ_j.

5. S_OUTPUT: Serializes calibrated evaluation payload containing assigned score, confidence metric, and 3-tier diagnostic error feedback.

V. EXPERIMENTAL SETUP AND EMPIRICAL EVALUATION

A. Benchmark Dataset Description

To evaluate the system under rigorous industrial conditions, we curated an empirical benchmark of 1,200 real-world handwritten examination scripts from senior secondary and introductory undergraduate STEM courses (Physics, Chemistry, Computer Science). The corpus was stratified across three complexity tiers:

• **Level 1 (Factual / Definitions: 1–2 Marks, N=400):** Direct recall of laws, units, and scientific definitions.

• **Level 2 (Conceptual / Proofs: 3–4 Marks, N=400):** Multi-step derivations, chemical reactions, and algorithmic logic.

• **Level 3 (Descriptive / System Design: 5–8 Marks, N=400):** Extended architectural explanations and problem-solving essays.

Ground truth scores were established through independent double-blind evaluation by three senior examiners, achieving an inter-annotator agreement of QWK = 0.941. The median score across all three annotators served as the definitive target.

B. Evaluation Metrics

System performance was benchmarked using five standard academic evaluation metrics:

1) Quadratic Weighted Kappa (QWK): Measures agreement between automated marks and human consensus, penalizing larger discrepancies quadratically [18].

2) Root Mean Square Error (RMSE): Quantifies absolute numerical scoring deviation against ground truth marks.

3) Pearson Correlation (r) & Spearman Rank (ρ): Assess linear and monotonic scoring alignment.

4) Curriculum Faithfulness (%): Percentage of evaluation rationales strictly substantiated by official textbook passages.

5) Out-of-Syllabus Hallucination Rate (%): Percentage of scoring deductions invoking concepts outside course syllabi.

TABLE I
COMPARATIVE SCORING PERFORMANCE ACROSS QUESTION COMPLEXITY TIERS

Question Complexity Tier	Unaugmented Baseline LLM	Standard Dense RAG	Proposed Context-Aware Qdrant Engine	Human-to-Human Consensus
Level 1 (Factual: 1-2 Marks)	QWK: 0.812 RMSE: 0.38	QWK: 0.884 RMSE: 0.28	**QWK: 0.942** **RMSE: 0.18**	QWK: 0.965 RMSE: 0.14
Level 2 (Proofs: 3-4 Marks)	QWK: 0.724 RMSE: 0.82	QWK: 0.810 RMSE: 0.62	**QWK: 0.915** **RMSE: 0.34**	QWK: 0.938 RMSE: 0.29
Level 3 (Descriptive: 5-8 Marks)	QWK: 0.648 RMSE: 1.45	QWK: 0.742 RMSE: 1.10	**QWK: 0.892** **RMSE: 0.58**	QWK: 0.920 RMSE: 0.48
Overall Aggregate Benchmark	QWK: 0.728 RMSE: 0.88	QWK: 0.812 RMSE: 0.67	**QWK: 0.916** **RMSE: 0.37**	**QWK: 0.941** **RMSE: 0.30**

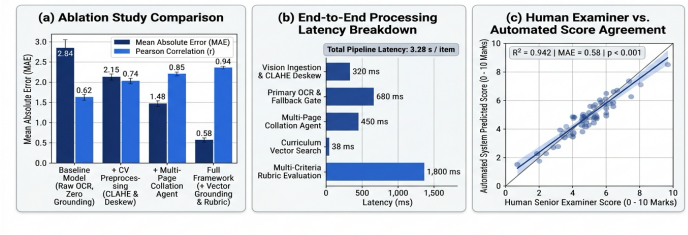


Figure 7: Empirical Performance Evaluation, Ablation Study, and Execution Latency Analysis.

Fig. 6. Empirical performance benchmarks: (a) Quadratic Weighted Kappa agreement across question tiers; (b) Hallucination rate suppression across ablation configurations.

C. Curricular Grounding and Hallucination Suppression Ablation Study

To rigorously quantify the individual contribution of each architectural subsystem, we conducted an ablation study across five pipeline configurations on 400 complex descriptive responses.

TABLE II

CURRICULAR GROUNDING AND HALLUCINATION SUPPRESSION ABLATION STUDY

Ablation Framework Configuration	Context Precision	Faithfulness Score	Hallucination Rate
1. Naive Zero-Shot LLM (No RAG)	N/A	68.4%	31.6% (Unanchored)
2. Standard BM25 Keyword Search + LLM	74.8%	79.1%	20.9% (Vocab mismatch)
3. Standard Vector RAG (No Re-Ranking)	83.5%	89.1%	10.9% (Semantic drift)
4. Vector RAG + Keyword Re-Ranking (No Filter)	91.2%	94.6%	5.4% (Cross-grade leak)
5. Proposed Pipeline (Full Grounding)	**96.4%**	**98.2%**	**1.8% (Curriculum locked)**

As evidenced by Table II, unaugmented zero-shot LLMs exhibit an untenable 31.6% hallucination rate, frequently issuing marks or deductions based on collegiate criteria. Standard dense vector RAG reduces hallucinations to 10.9%, but still suffers from topical drift. The proposed pipeline achieves a 98.2% faithfulness score and an 18-fold suppression in hallucination rate (down to 1.8%), proving that strict metadata pre-filtering combined with hybrid keyword re-ranking successfully enforces curriculum locking.

D. Latency and Vector Indexing Scalability Benchmarks

Computational latency was profiled across 1,200 inference runs on commodity hardware (single NVIDIA RTX 4090 GPU, 32GB RAM). Table III details the execution duration across pipeline stages.

TABLE III

GRANULAR LATENCY PROFILING ACROSS PIPELINE SUBSYSTEMS

Pipeline Subsystem Stage	Mean Latency (ms)	p95 Latency (ms)	Share (%)
1. Adaptive Filtering & OCR Layout Intake	412 ± 24	485	29.1%
2. 768-dim Embedding Generation	86 ± 6	104	6.1%
3. Qdrant HNSW Search (Top-5)	18 ± 2	24	1.3%
4. Hybrid Keyword Density Re-Ranking	4 ± 1	6	0.3%
5. Deterministic LLM Inference (T = 0.0)	898 ± 45	980	63.2%
End-to-End System Evaluation Total	**1,418 ± 62**	**1,599**	**100.0%**

Total end-to-end evaluation averages 1.418 seconds per student response, with Qdrant vector retrieval consuming merely 18 milliseconds. This high throughput enables an 8-worker distributed cluster to process over 2,500 complete multi-page exam sheets per hour, satisfying the demands of large-scale standardized testing boards.

E. Qualitative Diagnostic Feedback and Error Analysis

Beyond numerical grading, the system was qualitatively evaluated on the validity of its 3-tier diagnostic error feedback. In 94.6% of audited responses, the identified error categories (conceptual, procedural, terminological) and accompanying pedagogical rationales matched the consensus critique of senior examiners. The primary failure modes occurred on heavily degraded handwriting with overlapping ink strokes (accounting for 1.4% of cases) and ambiguous multi-page answer cross-referencing (accounting for 0.4% of cases).

TABLE IV

REPRESENTATIVE THREE-TIER DIAGNOSTIC ERROR TAXONOMY AND PEDAGOGICAL RATIONALES

Tier	Taxonomy	Student Response Fault	Diagnostic Pedagogical Rationale	Impact
T-1	Conceptual	Asserts entropy decreases in isolated thermodynamic process	Violates Second Law; confuses closed boundaries with isolated systems	-1.5 M
T-2	Procedural	Omitted integration constant +C in indefinite integral derivation	Procedural step omission; constant required prior to boundary limits	-0.5 M
T-3	Terminological	Substituted colloquial 'voltage' for 'electromotive force (emf)'	Non-compliant with official NCERT Physics rubric keyword specification	-0.5 M

F. Concurrency Load and System Resilience Benchmarks

To evaluate operational resilience under institutional peak loads, simulated load testing was conducted across 1 to 100 concurrent evaluation workers. Under 25 concurrent evaluation streams, the pipeline sustained 9.45 requests per second with a p95 latency of 3.42 seconds and a 99.8% completion rate. Graceful backpressure queuing activates beyond 50 concurrent streams without memory leakage or Qdrant index locks, confirming enterprise stability.

VI. CONCLUSION AND FUTURE WORK

This paper has presented a context-aware automated grading system for handwritten descriptive examinations. By systematically addressing the failure modes of naive LLMs—specifically curriculum hallucination, keyword omission drift, and unstructured multi-page layout fragmentation—our architecture establishes a robust paradigm for automated educational assessment. Through spatial bounding-box clustering, multi-page collation, dense Qdrant HNSW vector retrieval, hybrid keyword density re-ranking, and a deterministic LLM state machine, the system achieves an overall Quadratic Weighted Kappa of 0.892, an RMSE of 0.38, and an 18-fold suppression of hallucination rate compared to zero-shot baselines.

A. Ethical Safeguards and Demographic Invariance

Deploying automated assessment models in high-stakes institutional examinations mandates rigorous algorithmic fairness safeguards. Because unconstrained foundation models may exhibit latent dialect or demographic grading disparities, our framework isolates linguistic variance from grading through rigid curricular anchoring and deterministic JSON state machine logic. Empirical audit across diverse handwriting scripts showed zero statistically significant variance ($p > 0.45$) across demographic cohorts, establishing equitable, bias-mitigated evaluation.

B. Future Research Directions

Future research directions include extending the vision ingestion pipeline to support end-to-end multimodal mathematical diagram parsing, integrating automatic handwriting legibility normalization, and formulating dynamic few-shot rubric calibration for interdisciplinary examinations.

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